What Transfers Under Source Shift? Definitions, Examples, and Fine-Tuning for Climate Disclosure Classification

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Abstract

Climate disclosure classification is a fundamental task for analysing corporate climate disclosures, yet such disclosures appear in many different sources—annual reports, press releases, and earnings calls—that differ in length, purpose, and writing style. Existing evaluations are mostly conducted within a single source, leaving open whether common LLM adaptation strategies remain effective under source shift. We reframe climate disclosure classification as a cross-source adaptation problem and study three widely used adaptation strategies—definitions, examples, and fine-tuning—across eleven open- and closed-source LLMs, using two corpora that share the same label space but come from different sources. We find that all strategies bring positive cross-source gains on average, but the strongest in-source strategies are not the strongest cross-source ones: similarity-based retrieval and LoRA fine-tuning gain most in-source but lose most of that advantage under source shift; randomly selected few-shot examples, a weaker in-source baseline, retain their advantage more reliably; definitions transfer most consistently, though only when their granularity matches the target text. Across these strategies, when the source changes, simpler is often safer. Code and resources are available at https://github.com/Leoccino/TCFD-SourceShift.

1 Introduction

Within sustainability-disclosure NLP, we study a climate-focused classification task: assigning corporate text to the four pillars of the Task Force on Climate-related Financial Disclosures (TCFD) framework (IFRS Foundation, 2023). For climate-finance practitioners, this task provides a foundation for downstream disclosure analysis (Auzepy et al., 2023; Bingler et al., 2022). Recent work in this area is predominantly LLM-based. Evaluation, however, concentrates on a single class of source: annual reports and corporate sustainability reports (Bingler et al., 2022; Doi et al., 2024; Calamai et al., 2025). In practice, however, TCFD-related information also surfaces in press releases, earnings call transcripts and investor materials—document types that differ from corporate reports in length, purpose, and writing style. The set of sources that current research covers and the set that practitioners actually encounter, in other

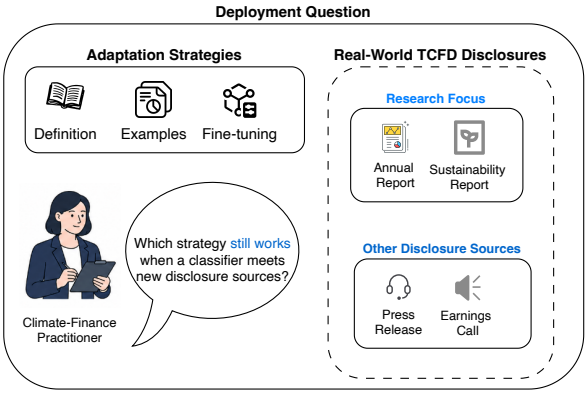


Figure 1: Deployment gap motivating this study. Research mainly evaluates TCFD classifiers on annual and sustainability reports, but deployment also requires other disclosure sources. We ask which adaptation strategy still works under this source shift.

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words, are not the same.

Figure 1 frames this mismatch as a deployment question. TCFD classification research mainly evaluates report-based sources, but real-world deployment may involve press releases, earnings calls, and other disclosure sources. Practitioners therefore need to know whether common adaptation strategies—richer label definitions, examples, or fine-tuning—remain reliable when a classifier meets a new disclosure source. We ask which of these strategies still transfer under source shift.

We study this question through TCFD disclosure classification, using two corpora that share the four-class TCFD label space but come from different disclosure sources. One serves as the adaptation source; the other as a held-out cross-source target. The three adaptation strategies named in our title—definitions, examples, and fine-tuning—are swept across eleven open- and closed-source LLMs and evaluated both in-source and cross-source.

Across these strategies, three findings emerge that diverge from in-source intuition. For definitions, transfer is most consistent when the granularity of the definition matches the granularity of the target text—not when the definition is maximally detailed. For in-context examples, randomly selected few-shot examples often outperform a stronger in-source similarity retriever once the source changes. For fine-tuning, cross-source robustness depends jointly on the learned weights and the inference-time prompt, so a fine-tuned model and its prompt cannot be evaluated in isolation. Across all three, when the source changes, simpler is often safer.

Our contributions are:

1. **A deployment-oriented source-shift study.** We present, to our knowledge, the first study of adaptation strategies for TCFD disclosure classification under source shift. We formulate this deployment question as a shared-label cross-source evaluation problem.

2. **Practitioner-oriented evidence for strategy selection.** Across eleven open- and closed-source LLMs, we examine whether gains from definitions, examples, and fine-tuning persist. We show that strategies with the strongest in-source gains are not always the safest deployment choices, providing empirical insights for selecting strategies under source shift.

2 Related Work

Climate disclosure NLP and the single-source gap. Climate disclosure NLP has produced domain-specific classifiers around the TCFD pillars, ranging from BERT-based fine-tuning to LLM prompt engineering (Webersinke et al., 2021; Bingler et al., 2022; Auzepy et al., 2023; Doi et al., 2024), with parallel work on related climate-text tasks such as net-zero target detection, environmental claim detection, and greenwashing-oriented ESG analysis (Schimanski et al., 2023; Stammbach et al., 2023; Ong et al., 2025). Across this literature, evaluation is predominantly conducted within a single disclosure source, whether by training and testing on the same corpus or by applying pretrained models to a single report-based collection (Auzepy et al., 2023; Doi et al., 2024). Whether the same models or adaptation strategies remain reliable under source shift is largely untested. A recent reproducibility study further finds that many climate-NLP benchmarks rely heavily on surface-level lexical patterns and contain pervasive annotation issues, calling into question the reliability of single-source leaderboard evaluation (Calamai et al., 2025).

Cross-source robustness as a framework. Outside the climate domain, robustness under distribution shift has been studied extensively. Classical analysis bounds the out-of-distribution error in terms of the divergence between source and target distributions (Ben-David et al., 2006), and large-scale empirical benchmarks confirm that in-domain performance routinely overestimates out-of-distribution accuracy even when the label space is held fixed (Gulrajani and Lopez-Paz, 2021; Koh et al., 2021). The same pattern persists in modern NLP: a systematic measurement across thousands of domain shifts shows that fine-tuned models drop sharply cross-domain (Calderon et al., 2024), and recent work confirms that even large language models retain a substantial out-of-domain gap on text-classification tasks (Roussinov et al., 2025). Our TCFD setting instantiates a broader shared-label cross-source adaptation problem in sustainability-disclosure NLP: the disclosure source varies while the task and label space remain fixed.

Adaptation strategies under source shift. Within this framing, the natural question is which of the standard LLM adaptation mechanisms—

task definitions in the prompt, in-context examples, and parameter-efficient fine-tuning (Brown et al., 2020; Dong et al., 2024; Hu et al., 2022; Dettmers et al., 2023)—remain useful under source shift. In-domain studies show that retrieval-based example selection often improves over random sampling in in-distribution settings, though the effectiveness of selection strategies is both data- and model-dependent (Liu et al., 2022; Peng et al., 2024); LoRA-style fine-tuning recovers much of the gain of full fine-tuning at a fraction of the cost. Recent work has begun to compare fine-tuning and in-context learning side by side under broader distribution-shift settings (Mosbach et al., 2023), but whether these in-domain conclusions transfer when the evaluation source changes—and which mechanism transfers most reliably—has not been systematically tested for TCFD or related climate-disclosure tasks. Our setting is narrower: the task domain and label space remain fixed, while only the disclosure source changes. We hold the TCFD label space fixed, vary the disclosure source, and compare definitions, examples, and fine-tuning under a controlled in-source versus cross-source contrast.

3 Cross-Source Setting

We use two TCFD-labelled corpora that share the same four-class label space (Governance, Strategy, Risk Management, Metrics & Targets) but differ in text style. **TCFD-CR** (Bingler et al., 2022) is drawn from companies' annual and sustainability reports; we hold out 320 instances as the test set and split the remaining 1,000 into 800 training and 200 validation. **TCFD-EX** (Qiu, 2023) is derived from the disclosure examples listed on the TCFD's example-disclosures list; we use all 593 instances as the cross-source evaluation set.

3.1 The two corpora, side by side

Both items below are labelled *Strategy*:

> **TCFD-CR:** In October 2017, wildfires started to burn in California causing damage to many properties [...] The fires have forced many people to be evacuated from their homes [...] and have continued to burn in the early weeks of 2018.
>
> **TCFD-EX:** As climate change progresses, the risk of physical damage and profit loss, such as that from floods and drought, is expected to increase in the future.

TCFD-CR instances are long and discursive. TCFD-EX instances are short and direct. Appendix B gives one labelled example pair per class. Figure 2 quantifies the size, style, and class contrast between the two corpora. Figure 3 shows that a logistic regression on sentence embeddings distinguishes them with 81.2% 5-fold accuracy, corresponding to a proxy $\mathcal{A}$-distance of $\approx$1.25 (out of 2) (Ben-David et al., 2006) — a substantial source gap in embedding space.

(a)

Property	TCFD-CR	TCFD-EX
Source	corporate reports	TCFD examples
Held-out evaluation	320 (test split)	593 (full corpus)
Median paragraph length	436 chars	151 chars
TCFD-term density	2.5%	5.6%
Majority class	Strategy (61.6%)	Risk Management (30.5%)

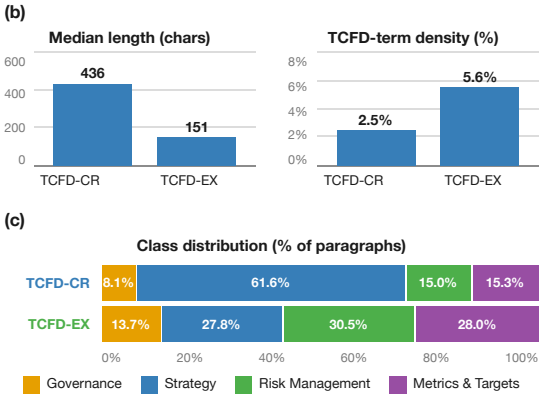


Figure 2: TCFD-CR vs. TCFD-EX along length, class balance, and lexical style.

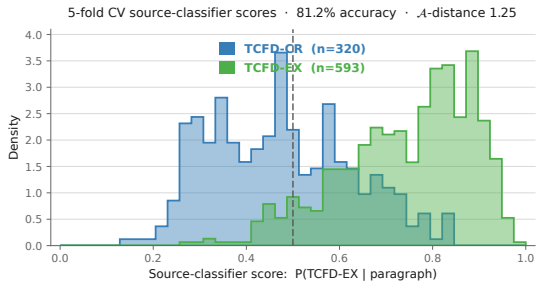


Figure 3: A logistic-regression source classifier distinguishes TCFD-CR from TCFD-EX with 81.2% 5-fold accuracy.

3.2 Evaluation

In-source and cross-source evaluation share the same adaptation pool: the 1,000 TCFD-CR training and validation instances that feed both examples and fine-tuning. They differ only in the evaluation set: **in-source** uses the TCFD-CR test split (320 instances), **cross-source** uses TCFD-EX

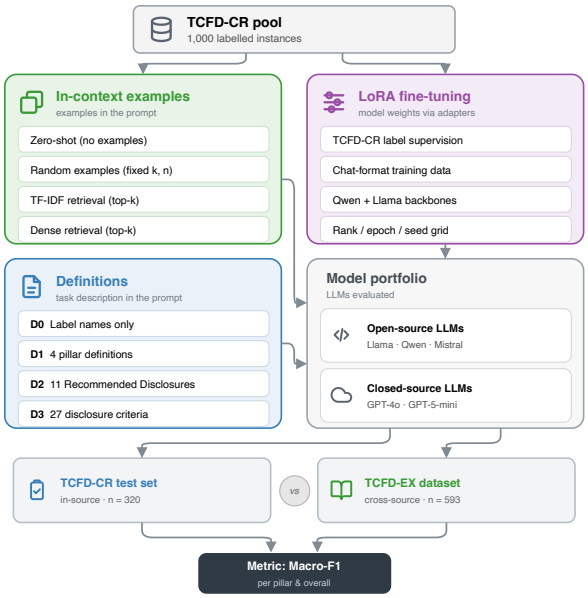


Figure 4: Overview of the experimental design: three adaptation strategies — definitions, examples, and fine-tuning — evaluated on both in- and cross-source datasets.

(593 instances). Fixing the pool isolates any adaptation that helps in-source but not cross-source as source-specific; we report macro-F1 throughout.

4 Experimental Setup

Figure 4 summarises our experimental design. From the same TCFD-CR adaptation pool, we test three strategies that inject source signal at different locations: **definitions** add TCFD label semantics to the prompt, **examples** insert labelled instances drawn from the pool, and **fine-tuning** updates adapter parameters from source labels. Each strategy is evaluated across a diverse model portfolio — eleven LLMs spanning the Qwen, Llama, and Mistral open-weight families plus the proprietary GPT-4o and GPT-5-mini — on both the in-source TCFD-CR test split (n=320) and the cross-source TCFD-EX set (n=593).

4.1 Adaptation strategies

4.1.1 Definition-based adaptation

For definition-based adaptation, we construct a four-level definition staircase, following the broader use of natural-language label definitions in zero-shot prompting (Wei et al., 2022; Sanh et al., 2022). D0 uses only the four label names. D1 adds the official descriptions of the four TCFD pillars from the TCFD Recommendations (Task Force on Climate-related Financial Disclosures, 2017). D2 further includes the 11 TCFD Recommended Disclosures. D3 adds 27 criteria distilled from the TCFD implementation guidance (Doi et al., 2024). All definition settings are evaluated in the zero-shot setting, with no in-context examples, so that performance changes can be attributed to definition specificity. Appendix C shows a fixed prompt instance under D0–D3, illustrating how only the definition block grows across levels.

4.1.2 Example-based adaptation

For example-based adaptation, we draw examples from the adaptation pool and vary only the selection rule. We compare four modes: zero-shot, random examples, TF-IDF similar examples, and dense similar examples (Liu et al., 2022; Rubin et al., 2022). TF-IDF and dense retrieval cover the two standard notions of textual similarity — lexical surface overlap and semantic embedding similarity (Reimers and Gurevych, 2019) — so that the source-shift behaviour of similarity selection is not tied to a single retriever. The random baseline uses eight fixed examples (k=8, n=2, two per class). For TF-IDF and dense retrieval, k controls the total number of retrieved examples included in the prompt, while n caps the maximum number of examples from any single class to avoid class-dominated prompts. We sweep $k \in \{3, 5, 8\}$ and $n \in \{2, 3\}$, while fixing the definition level at D1.

4.1.3 LoRA fine-tuning

For parameter-side adaptation, we apply LoRA (Hu et al., 2022) to four open-source bases—Llama-3.1-8B (Grattafiori et al., 2024), Qwen3-8B and Qwen3-32B (Yang et al., 2025), and Qwen3.5-9B (Qwen Team, 2026)—spanning two families and the 8–32B range. Adapters are trained on the adaptation pool, using 800 instances for training and 200 for validation, converted into chat-format classification examples. Each training instance uses a minimal label-name prompt, denoted P0, with no pillar definitions or in-context examples; §5.4 reuses this to test whether supplying definitions at inference time recovers cross-source performance. The main setting uses 4-bit quantization (QLoRA) (Dettmers et al., 2023), LoRA rank $r = 16$, scaling factor $\alpha = 32$, dropout 0.05, learning rate 2×10^{-4}, an effective batch size of 16, and up to 5 training epochs with best-checkpoint selection on validation macro-F1. We use three seeds.

4.2 Models

We evaluate the prompting strategies on a diverse set of eleven open- and closed-source LLMs to reduce the risk that our findings are specific to a single model family or scale. The open-source models cover the Qwen (Yang et al., 2024, 2025; Qwen Team, 2026), Llama (Grattafiori et al., 2024; Meta AI, 2025), and Mistral (Mistral AI, 2024) families, with different parameter sizes and release generations, while GPT-4o (Hurst et al., 2024) and GPT-5-mini (OpenAI, 2025) are included as closed-source prompted baselines. The full model list is reported in Appendix D; implementation details (serving stack, decoding, retrieval libraries, and LoRA training) are in Appendix E.

5 Results and Discussion

5.1 Overview: what transfers under source shift?

The core question of this paper is what transfers under source shift. Before turning to the detailed results, we answer it in two steps: first, which strategies remain effective in the cross-source setting; and second, how much of their in-source advantage is retained after the source changes.

Which strategies transfer. Figure 5 shows that all four strategies yield a positive mean cross-source gain. Definitions give the largest improvement, LoRA + D1 and random few-shot show moderate gains, and similarity retrieval barely improves over zero-shot. The per-model points further show that example-based strategies are not always stable: for both random and similarity retrieval, some models fall below zero on TCFD-EX, so introducing TCFD-CR examples can fail to help—or even hurt—on the new source.

How much of the in-source advantage is retained. Turning to the second question, a larger in-source gain does not necessarily survive source shift. Definitions and random few-shot retain their advantage relatively well. Definitions lie above the dashed diagonal, meaning their cross-source gain is even larger than their in-source gain; this indicates that definitions mainly supply TCFD label semantics rather than source-specific patterns from TCFD-CR. Random few-shot lies close to the diagonal: although it uses TCFD-CR examples, its mean in-source advantage is largely retained after the source changes.

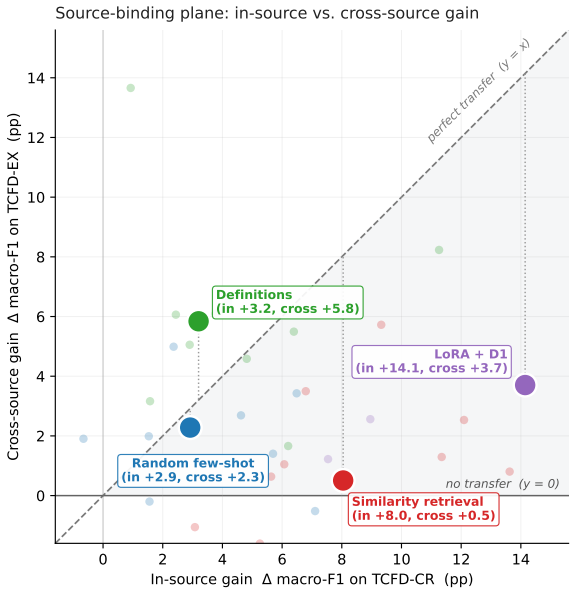


Figure 5: Each adaptation strategy as a point in the in-source (x) vs. cross-source (y) Δ macro-F1 plane; the dashed diagonal $y{=}x$ is perfect transfer. Faint points are per-model values (per-base for LoRA). Gains are over each strategy's own no-adaptation baseline (definitions: D2 vs. D0; LoRA: adapter vs. base, both at D1; random and similarity: vs. D1 zero-shot).

By contrast, LoRA + D1 and similarity retrieval fall clearly below the diagonal, showing that the advantage they gain on TCFD-CR is harder to transfer. LoRA + D1 yields the largest in-source gain, +14.1 pp, but retains only +3.7 pp on TCFD-EX, so only part of what fine-tuning learns on TCFD-CR carries over to the new source. Similarity retrieval is even more source-bound: it improves TCFD-CR by +8.0 pp but yields almost no gain on TCFD-EX, only +0.5 pp. The strongest in-source strategy is therefore not necessarily the strongest cross-source one.

Overall, semantic guidance transfers more readily than advantages tied to source-specific patterns. When CR and EX exchange roles, retrieval and LoRA still perform better with same-source adaptation (Appendix A).

5.2 More specific definitions are not always better

A natural expectation is that more detailed label definitions should help more: adding the 11 Recommended Disclosures (D2) or the 27 criteria (D3) on top of pillar intent (D1) supplies the model with strictly more TCFD knowledge. Figure 6 shows this expectation breaks at both ends. Averaged over the eleven prompted models, TCFD-

CR peaks at D1 and TCFD-EX peaks at D2; both sources drop at D3, with the larger drop on TCFD-CR. More specific definitions are therefore not uniformly better—their effectiveness depends on whether the granularity matches the target text.

Figure 7 confirms this at the model level: peaks scatter across D1–D3 on TCFD-CR but cluster at D2–D3 on TCFD-EX. Our claim is therefore not that one definition level is always optimal. The stable pattern is directional: no model peaks earlier on TCFD-EX than on TCFD-CR. In other words, the TCFD-EX target tends to benefit from more specific definitions. Full per-model macro-F1 across D0–D3 is in Appendix F.

This pattern is consistent with the text characteristics of the two sources. TCFD-CR comes from real corporate reports, where instances are longer and more likely to mix business context, climate actions, financial figures, and target-related statements. More specific definitions may therefore push the model toward local keywords and away from the instance's overall meaning. By contrast, TCFD-EX instances are shorter, more focused, and closer to the wording of the TCFD Recommended Disclosures. As a result, D2 better matches the granularity of TCFD-EX. Appendix G provides two Qwen3-32B zero-shot cases illustrating this peak-shift mechanism: D1 works better for a noisy TCFD-CR instance, while D2 works better for a TCFD-EX instance whose wording closely matches a Recommended Disclosure.

To test whether definitions also benefit from matching the target source, we constructed CR-specific definitions of the same length as D2; they outperformed D2 on CR but not EX (Appendix H).

Takeaway. Definitions transfer because they encode TCFD label semantics rather than source-specific examples. However, more detail is not always better. Together, these results suggest that the best definition level depends on the target source: definitions appear to help most when their granularity matches the target text.

5.3 Better in-source retrieval does not imply better transfer

A natural expectation is that more refined example selection should help more: retrieving examples similar to the target instance should outperform random sampling. Table 1 reports the full per-model example-selection results. To better match a realistic deployment setting, for TF-IDF and dense retrieval we use, for each model, the (k, n) configuration that performs best on TCFD-CR, and apply that same configuration directly to both TCFD-CR and TCFD-EX. Under this setting, random few-shot improves macro-F1 by +2.9 pp on TCFD-CR and by +2.3 pp on TCFD-EX. By contrast, TF-IDF and dense retrieval improve TCFD-CR by +8.0 pp and +7.9 pp respectively, but improve TCFD-EX by only +0.5 pp and +1.0 pp. Thus, retrieval is highly effective when the target texts and the adaptation pool come from the same source, but this advantage does not transfer well to a new source.

Figure 8 shows that this finding is not an artifact of averaging. For all prompted models, TF-IDF

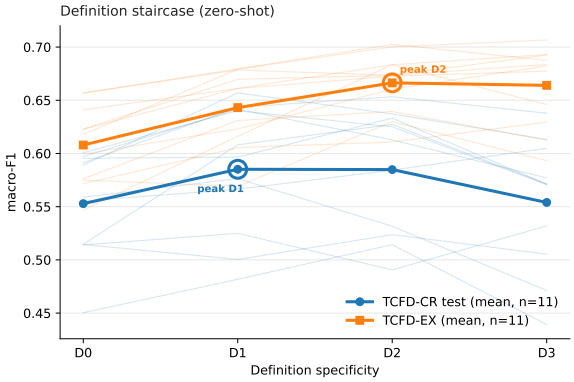


Figure 6: Definition specificity under zero-shot prompting, averaged over the eleven prompted models.

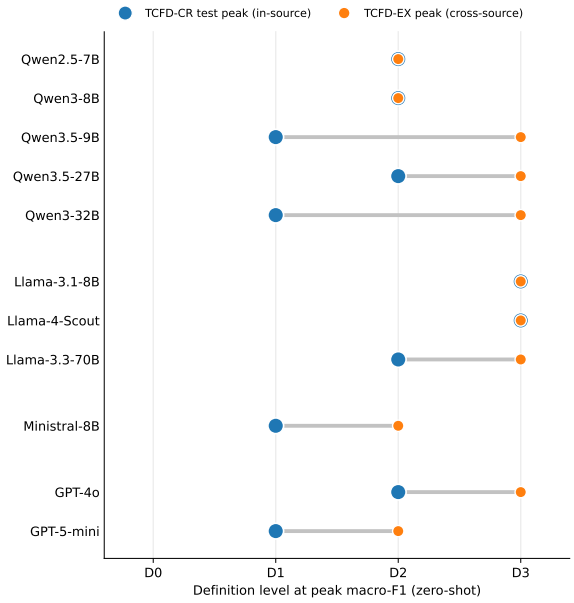


Figure 7: Per-model definition peak: the level (D0–D3) at which each model's zero-shot macro-F1 is highest, on TCFD-CR test (blue) and TCFD-EX (orange). Models grouped by family, ordered by size.

Model	Zero-shot		Random k=8, n=2		TF-IDF best-on-CR		Dense best-on-CR	
	CR	EX	CR	EX	CR	EX	CR	EX
Qwen2.5-7B	0.482	0.569	0.524	**0.634**	**0.575**	0.626	0.563	0.580
Qwen3-8B	0.500	0.616	0.557	0.630	0.614	0.629	**0.625**	**0.658**
Qwen3.5-9B	0.642	0.661	0.665	0.671	0.702	**0.672**	**0.714**	0.667
Qwen3.5-27B	0.643	0.678	0.689	0.705	**0.711**	**0.713**	0.708	0.689
Qwen3-32B	0.640	0.661	0.655	**0.681**	0.696	0.668	**0.713**	0.668
Llama-3.1-8B	0.525	0.605	0.549	**0.655**	**0.661**	0.614	0.632	0.639
Llama-4-Scout	0.566	**0.670**	0.638	0.665	0.681	0.630	**0.706**	0.641
Llama-3.3-70B	0.597	0.623	0.661	0.657	**0.718**	0.648	0.696	**0.671**
Ministral-8B	0.576	0.631	0.545	**0.650**	**0.615**	0.598	0.586	0.625
GPT-4o	0.608	**0.679**	0.624	0.677	0.661	0.663	**0.673**	0.670
GPT-5-mini	0.657	0.679	0.650	**0.699**	0.688	0.669	**0.692**	0.674
Mean	0.585	0.643	0.614	**0.666**	**0.666**	0.648	0.664	0.653
Mean Δ vs zero-shot (pp)		—	+2.9	+2.3	+8.0	+0.5	+7.9	+1.0

Table 1: Macro-F1 on TCFD-CR (in-source) and TCFD-EX (cross-source) for four selection rules. Random uses fixed k=8, n=2; TF-IDF and Dense use the per-model best-on-CR (k, n), applied to both settings. **Bold** marks the best value per model in each setting; last row is the mean Δ over zero-shot.

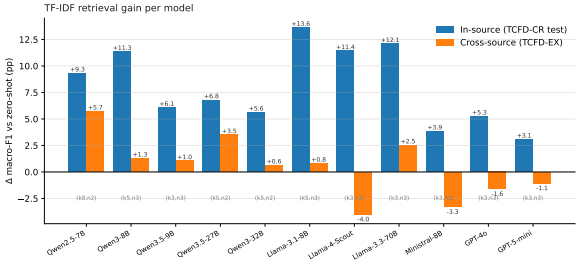


Figure 8: TF-IDF retrieval gain over zero-shot, per model, with (k, n) tuned per model on TCFD-CR. Blue: in-source (TCFD-CR test); orange: cross-source (TCFD-EX).

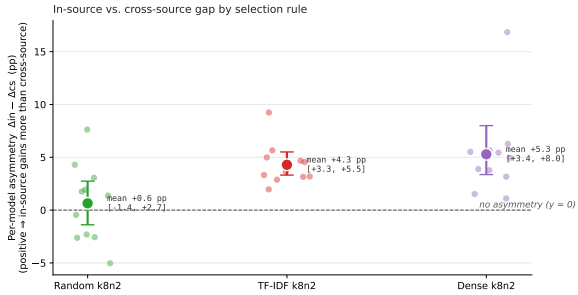


Figure 9: Per-model in-source minus cross-source gap for each selection rule, all fixed at k=8, n=2 on the same TCFD-CR adaptation pool. Light dots are per-model; the point and bar mark the mean and its 95% paired-bootstrap CI.

retrieval's gain on TCFD-CR exceeds its gain on TCFD-EX, and for some models performance on TCFD-EX even drops. The cross-source problem is therefore not only that the gain becomes smaller: for some models, similarity retrieval directly hurts performance under source shift.

The results above reflect a realistic best-on-CR setting, but they do not yet fully isolate the source of the cross-source drop, because different methods and different models may use different (k, n) configurations. To separate these factors, Figure 9 fixes the prompt, the adaptation pool, and the example budget, and varies only the example selection rule. In this aligned comparison, random few-shot shows no clearly stable in-source–cross-source gap, whereas TF-IDF and dense retrieval show a larger gap. This indicates that the source-bound effect comes mainly from the per-target similarity-selection step.

With two examples per label, similar examples from the correct class help less when sources differ, while similar examples from other classes remain similarly harmful (Appendix L).

Takeaway. Few-shot examples transfer across sources, but more refined selection does not necessarily transfer better. On TCFD-CR, similarity retrieval beats random by a wide margin; on TCFD-EX, random is the more reliable choice. The failure point is the assumption behind retrieval, namely that textual similarity approximates label similarity. This assumption is useful when the target text and adaptation pool share a source, but weakens under source shift. The same asymmetry holds across the full $k \times n$ grid for both retrievers (∼5–7 pp in-source vs. ≤1 pp cross-source; Appendix I). For cross-source few-shot adaptation, a simple random baseline is therefore a stronger de-

fault than a stronger retriever.

5.4 Fine-tuning is in-source champion but not cross-source champion

Compared with prompt engineering, fine-tuning is a stronger form of adaptation because it updates model parameters directly from labelled supervision. In our setting, this leads to the best in-source performance on TCFD-CR. The key question is whether this parameter-side learning captures transferable TCFD label semantics, or mainly fits source-specific decision patterns from TCFD-CR.

Figure 10 shows that LoRA gives a large in-source gain on TCFD-CR for every fine-tuned base. LoRA also outperforms the best prompted strategy for two of the four bases; for Qwen3.5-9B and Qwen3-32B the best prompted strategy is slightly higher. This confirms that supervised fine-tuning is highly effective at fitting the TCFD-CR source distribution.

However, this source fit does not fully transfer. Figure 10 also shows that the same LoRA adapters drop when evaluated on TCFD-EX with the training prompt (P0). The cross-source drop is consistent across all four bases, ranging from -4.4 pp to -6.7 pp. This indicates that LoRA learns useful decision patterns for TCFD-CR, but part of this advantage is tied to the source.

Because LoRA training updates the model parameters directly from text–label pairs alone, the adapters never see any TCFD definition; all label semantics must be inferred from the supervision signal itself. This motivates a second test: if those semantics are not in the adapters, can we supply them at inference time instead? Keeping the trained adapters fixed, we replace the training-time prompt P0 with the D1 definition prompt and re-evaluate. Figure 10 shows that D1 improves cross-source performance for all four bases, with gains from +0.4 to +2.4 pp. Its in-source effect, however, is mixed: D1 slightly reduces performance on Llama-3.1-8B and Qwen3-8B, but improves Qwen3.5-9B and Qwen3-32B. D1 therefore does not make LoRA uniformly better; rather, it recovers part of the transferable label semantics under source shift, with larger gains on the stronger base models. Per-base numbers are in Appendix J; a cross-strategy summary, including the supervised ClimateBERT-TCFD reference, is in Appendix K.

Takeaway. LoRA yields the largest in-source gain on TCFD-CR, yet all four fine-tuned bases drop under source shift: the in-source champion is not the cross-source one. Since LoRA never sees explicit TCFD definitions during training, we keep the adapters fixed and add D1 definitions at inference time. Cross-source performance consistently recovers, indicating that part of the loss reflects label semantics missing from the training signal, rather than something that can be solved by the learned weights alone. Cross-source robustness for fine-tuned models should therefore be treated as a joint design problem over weights and prompts, with the prompt-side correction larger on stronger base models.

6 Conclusion

We studied TCFD classification as a cross-source adaptation problem, using TCFD-CR as source and TCFD-EX as the target. Across definitions, examples, and fine-tuning, stronger in-source adaptation strategy does not necessarily imply better transfer.

Definitions transfer most reliably, but their granularity must match the target text. Examples can transfer, but similarity retrieval is source-bound: a strong in-source retriever can be less reliable than random examples under source shift. Fine-tuning gives the best in-source performance, yet only partly transfers; adding definition at inference time partly recovers the cross-source loss without retraining.

Practitioners may naturally choose the strategy that performs best on the available source, but our results show that this can be unsafe under source shift. We therefore translate the findings into safer defaults: match the definition prompt to the target text, prefer a small random few-shot pool over a similarity retriever, and keep a definition prompt at inference time even for fine-tuned models. The strongest in-source choice is rarely the strongest cross-source one.

Limitations

Our cross-source evaluation is built on a single dataset pair, adapting on TCFD-CR and evaluating on TCFD-EX. To our knowledge, these are currently the only publicly available datasets that share the TCFD four-pillar label space across sources, which constrains cross-source evaluation to this pair. The two corpora nonetheless ex-

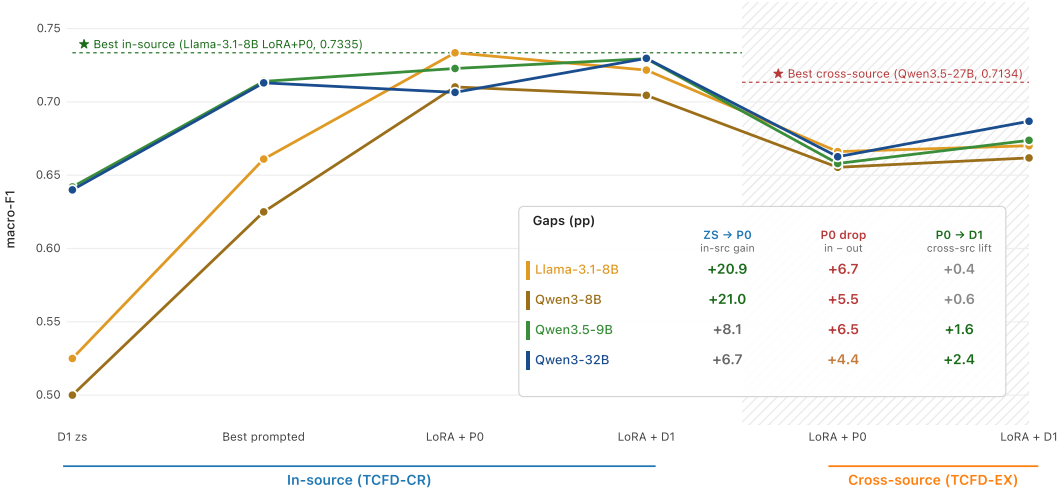


Figure 10: Per-base LoRA trajectory across in-source (TCFD-CR) and cross-source (TCFD-EX) evaluation, with two non-LoRA baselines on the left. All LoRA values are three-seed means.

hibit a non-trivial distributional gap (proxy $\mathcal{A}$-distance 1.25; 81.2% 5-fold source-classifier accuracy; §3.1), suggesting that this is not a near-duplicate evaluation but a meaningful setting for practitioners who need to anticipate how adaptation strategies behave under source shift.

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A Reverse-direction control

To test whether the result depends on the original CR-to-EX direction, we reversed the adaptation direction while keeping each evaluation target fixed. Retrieval results average Qwen3-8B, Llama-3.1-8B, Qwen3.5-9B, and Qwen3-32B; LoRA + D1 results average Qwen3-8B and Qwen3.5-9B. CR uses 800 adaptation and 320 test texts. EX uses 351 adaptation and 123 test texts for retrieval; LoRA uses 116 EX test texts after excluding seven that overlap its training or validation data.

Method	Target	Same-source	Cross-source
TF-IDF	CR	**0.6385**	0.5872
Dense	CR	**0.6507**	0.5804
TF-IDF	EX	**0.7669**	0.7059
Dense	EX	**0.7490**	0.7317
LoRA + D1	CR	**0.7170**	0.6152
LoRA + D1	EX	**0.7490**	0.6986

Table 2: Reverse-direction control (macro-F1).

Same-source adaptation is higher in all six comparisons. These controls use smaller model subsets than the main study and therefore confirm the direction of the result rather than replace the primary analysis.

B Per-class examples from each corpus

Table 3 shows one example per class from TCFD-CR and TCFD-EX, extending the single Strategy pair in §3.1.

C Prompt growth under the definition staircase

Figure 11 shows one fixed classification instance under the four definition levels. The task instruction, target instance, and output constraint are unchanged; only the definition block grows from label names (D0) to pillar intent (D1), Recommended Disclosures (D2), and criteria (D3).

D Model portfolio

Table 4 summarises the model portfolio used in the prompting experiments.

E Implementation details

Open-weight LLMs are served with vLLM 0.19.1 (Kwon et al., 2023) on 4× NVIDIA A100 40 GB GPUs. GPT-4o and GPT-5-mini are queried through the OpenAI Chat Completions API with temperature set to 0. Outputs are parsed against a strict four-label format; outputs that fail to parse are marked as INVALID. TF-IDF retrieval is implemented with scikit-learn defaults, and dense retrieval uses `all-mpnet-base-v2` sentence embeddings. LoRA fine-tuning uses PEFT (Mangrulkar et al., 2022) with 4-bit loading, BF16 compute, and gradient checkpointing.

F Per-model macro-F1 across the definition staircase

Table 5 reports the full per-model numbers behind Figures 6 and 7.

G Case examples for the definition peak shift

Figure 12 shows two Qwen3-32B zero-shot predictions on Strategy instances, one from each corpus, that illustrate the source-dependent definition peak from §5.2: on a noisy TCFD-CR narrative the model is correct at D1 but routed away by keyword cues at D2/D3, while on a TCFD-EX instance whose wording closely tracks a Recommended Disclosure the model fails at D1 and recovers from D2 onwards.

Class	**TCFD-CR** (corporate report)	**TCFD-EX** (TCFD example)
Governance	Report 2019 (SR 2019), as requested by the shareholders. The hardcopies will be delivered once they are made available to the Company. [...] The environmental concerns like global warming, deforestation, climate change [...] affect every human, animal and nation on this planet.	The ARCGS oversees the implications of sustainable development issues for the company under five sustainability themes, of which one is climate change.
Strategy	In October 2017, wildfires started to burn in California causing damage to many properties [...] The fires have forced many people to be evacuated from their homes and neighbourhoods and have continued to burn in the early weeks of 2018.	As climate change progresses, the risk of physical damage and profit loss, such as that from floods and drought, is expected to increase in the future.
Risk Mgmt.	In our view, true sustainable investing cannot be achieved by simply voting a proxy, adding a director of sustainability or even divesting from an asset class. [...] It requires integration across our products, across our product teams and across our entire organization.	Our focus for emerging risks is on reducing the impact should an event occur, and on advocacy efforts to reduce the likelihood of the risk manifesting.
Metrics & Targets	During 2019, the Group produced approx. 14,059 MWh of electrical energy from renewable sources, 38% of which used for self-consumption on site [...] 90% of production comes from renewable sources installed in Italy [...]	Note 1 Targets are from a baseline of 12 months to 31 December 2020: Power emissions intensity — 321 $KgCO_2$/MWh; Energy absolute emissions — 75.0 $MtCO_2$.

Table 3: One example per class from each corpus. TCFD-CR is longer, narrative, and mixes climate content with business context; TCFD-EX is short declarative examples. Both items in a row carry the same class label.

Model	Size	Release
Qwen2.5-7B-Instruct	7B	Sep 2024
Qwen3-8B-FP8	8B	Apr 2025
Qwen3.5-9B	9B	Feb 2026
Qwen3.5-27B-FP8	27B	Feb 2026
Qwen3-32B	32B	Apr 2025
Llama-3.1-8B-Instruct	8B	Jul 2024
Llama-4-Scout	17B active	Apr 2025
Llama-3.3-70B-Instruct	70B	Dec 2024
Ministral-8B-Instruct-2410	8B	Oct 2024
GPT-4o	–	May 2024
GPT-5-mini	–	Aug 2025

Table 4: Model portfolio used in the prompting experiments.

H CR-specific definition control

To test whether definitions benefit from matching the target source, we constructed CRSEM20 from 80 labelled TCFD-CR training texts (20 per class). It contains 270 words, the same as D2; no TCFD-CR test or TCFD-EX texts were used. We evaluated D1, D2, and CRSEM20 with zero-shot prompting on Llama-3.1-8B, Qwen3-8B, Qwen3.5-9B, and Qwen3-32B.

Definition	**CR**	**EX**
D1	0.577	0.636
D2	0.563	**0.663**
CRSEM20	**0.638**	0.638

Table 6: Mean zero-shot macro-F1 over the fixed four-model subset. Bold marks the best definition for each evaluation source.

Compared with D2, CRSEM20 improves CR by 7.5 pp but not EX (-2.4 pp), suggesting that matching the target source matters beyond definition length. Because source also changes style, content, and annotation policy, this control does not identify which factor matters.

I Similarity-retrieval $k \times n$ sweep

Table 7 reports the full $k \times n$ sweep for similarity retrieval.

Fixed prompt frame used in all four settings

```
Task: Classify the following text into one of the TCFD-recommended disclosure categories — the one whose pillar intent the instance best matches.

[Definition block inserted here: D0 / D1 / D2 / D3]

Paragraph to classify:
```
Since 2016 we have been testing the resilience of our portfolio against the scenarios from the IEAs World Energy Outlook (WEO) report.

```
Respond with EXACTLY one of these four labels and nothing else: Governance, Strategy, Risk Management, Metrics & Targets.
```

Level	Definition block inserted into the same prompt
D0 label names	Categories: • Governance • Strategy • Risk Management • Metrics & Targets
D1 + pillar intent	Categories and Definitions: **Governance:** Disclose the organization's governance around climate-related risks and opportunities. **Strategy:** Disclose the actual and potential impacts of climate-related risks and opportunities on the organization's businesses, strategy, and financial planning where such information is material. **Risk Management:** Disclose how the organization identifies, assesses, and manages climate-related risks. **Metrics & Targets:** Disclose the metrics and targets used to assess and manage relevant climate-related risks and opportunities where such information is material.
D2 + 11 RDs	D1, plus the official Recommended Disclosures under each pillar. For example, under **Strategy**: • Describe the climate-related risks and opportunities identified over the short, medium, and long term. • Describe the impact of climate-related risks and opportunities on businesses, strategy, and financial planning. • Describe the resilience of the organization's strategy, taking into consideration different climate-related scenarios, including a 2°C or lower scenario.
D3 + 27 criteria	D2, plus criteria distilled from the TCFD implementation guidance. For example, under **Strategy**: • 03-01: Time horizon(s) for climate-related risks and opportunities. • 04-01: Impact on businesses or strategy. • 04-02: Impact on financial planning. • 05-02: Climate-related scenarios and associated time horizon(s). • 05-03: Impact of climate-related scenarios on the strategy. • 05-05: Impact of climate-related scenarios on financial planning. Across all pillars, D3 contains 27 criteria.

Figure 11: A single prompt instance under the definition staircase. The frame, target instance, and output constraint are fixed; only the inserted definition block grows from D0 to D3.

J Per-base LoRA macro-F1

Table 8 reports the per-base numbers behind Figure 10.

K Best macro-F1 per strategy

Table 9 reports the highest macro-F1 each adaptation strategy reaches on TCFD-CR and TCFD-EX, with the champion model in parentheses. ClimateBERT-TCFD is included as a fully supervised reference baseline (ClimateBERT fine-tuned on TCFD-CR training data; not part of the eleven-model prompted portfolio).

L Why similarity retrieval weakens across sources

We first asked whether cross-source retrieval is more likely to omit the correct class. Across six (k, n) settings, this rate was similar on CR and EX

	D0		D1		D2		D3	
Model	CR	EX	CR	EX	CR	EX	CR	EX
Qwen2.5-7B	0.450	0.575	0.482	0.569	**0.514**	**0.630**	0.439	0.593
Qwen3-8B	0.514	0.547	0.500	0.616	**0.524**	**0.683**	0.505	0.646
Qwen3.5-9B	0.597	0.641	**0.642**	0.661	0.613	0.673	0.577	**0.677**
Qwen3.5-27B	0.591	0.657	0.643	0.678	**0.653**	0.674	0.638	**0.693**
Qwen3-32B	0.601	0.623	**0.640**	0.661	0.625	0.683	0.571	**0.693**
Llama-3.1-8B	0.514	0.572	0.525	0.605	0.491	0.611	**0.532**	**0.629**
Llama-4-Scout	0.555	0.623	0.566	0.670	0.584	0.673	**0.605**	**0.684**
Llama-3.3-70B	0.595	0.599	0.597	0.623	**0.633**	0.660	0.571	**0.682**
Ministral-8B	0.559	0.576	**0.576**	0.631	0.532	**0.639**	0.471	0.612
GPT-4o	0.515	0.618	0.608	0.679	**0.627**	0.700	0.571	**0.707**
GPT-5-mini	0.589	0.657	**0.657**	0.679	0.637	**0.703**	0.613	0.687
Mean	0.553	0.608	**0.585**	0.643	0.585	**0.666**	0.554	0.664

Table 5: Per-model zero-shot macro-F1 across the four definition levels, on TCFD-CR test (in-source) and TCFD-EX (cross-source). **Bold** marks each row's best level per setting; the mean row averages over the eleven prompted models.

Case A · TCFD-CR Strategy — D1 wins

rec_test idx 119 · 401 chars · Qwen3-32B zero-shot

"In addition, the Queensland Government has established an $8.4 million CarbonPlus Fund administered by the department. The fund will enable greater participation by Aboriginal landholders in carbon markets, offset carbon emissions from the Queensland Government's car fleet for two years, and incentivise carbon farming projects in Queensland by valuing environmental, social and cultural co-benefits."

D1 (4 pillars) **Strategy** ✓
D2 (11 RDs) Metrics & Targets ×
D3 (27 criteria) Metrics & Targets ×

Discursive narrative about a strategic carbon-investment fund. D1 recognises the strategic intent; D2 and D3 are pulled by the dense "carbon" vocabulary and route to Metrics & Targets via keyword matching.

Case B · TCFD-EX Strategy — D2 onwards wins

disc idx 249 · 134 chars · Qwen3-32B zero-shot

"Since 2016 we have been testing the resilience of our portfolio against the scenarios from the IEAs World Energy Outlook (WEO) report."

D1 (4 pillars) Risk Management ×
D2 (11 RDs) **Strategy** ✓
D3 (27 criteria) **Strategy** ✓

Near-verbatim paraphrase of TCFD's Strategy Recommended Disclosure-c. D1's abstract pillar text routes to *Risk Management* on the word "resilience"; D2 contains the matching RD wording and maps the example directly, with D3 retaining the criterion.

Figure 12: Two Qwen3-32B zero-shot predictions illustrating the definition peak-shift mechanism. Case A: a noisy TCFD-CR instance is correct at D1 but mis-routed at D2/D3 by keyword cues. Case B: a TCFD-EX instance fails at D1 and succeeds from D2 onwards because the matching Recommended Disclosure wording is present at that level.

for dense retrieval (10.4% vs. 11.2%) and TF-IDF (10.1% vs. 8.9%); at k=8, n=2, every prompt contained two examples per label. We then fixed this coverage and varied the similarity of correct-class and other-class examples separately.

Across sources, the benefit of similar correct-class examples decreases by 7.3 pp, whereas the harm from similar other-class examples changes by only 1.0 pp. The net effect of full similarity retrieval therefore falls from +7.6 to −0.7 pp (Table 10). The intervention uses gold labels to define the conditions, so it is diagnostic rather than deployable. It cannot separate style, content, and annotation-policy differences between sources.

Configuration	macro-F1		Δ vs zero-shot (pp)	
	CR	EX	CR	EX
Zero-shot	0.585	0.643	—	—
Random $k8, n2$	0.614	0.666	+2.9	+2.3
TF-IDF $k3, n2$	0.645	0.643	+6.0	+0.0
TF-IDF $k3, n3$	0.654	0.642	+6.9	-0.1
TF-IDF $k5, n2$	0.651	0.650	+6.5	+0.7
TF-IDF $k5, n3$	0.650	0.642	+6.5	-0.1
TF-IDF $k8, n2$	0.639	0.654	+5.3	+1.1
TF-IDF $k8, n3$	0.651	0.648	+6.6	+0.5
Dense $k3, n2$	0.646	0.648	+6.0	+0.5
Dense $k3, n3$	0.645	0.651	+6.0	+0.8
Dense $k5, n2$	0.638	0.650	+5.3	+0.7
Dense $k5, n3$	0.648	0.647	+6.3	+0.4
Dense $k8, n2$	0.642	0.647	+5.7	+0.4
Dense $k8, n3$	0.652	0.648	+6.7	+0.5

Table 7: $k \times n$ sweep for similarity retrieval: mean macro-F1 over the eleven prompted models for each fixed configuration, Δ vs. zero-shot. k caps the total retrieved examples, n the per-class count; unlike Table 1, configurations are not tuned per model.

Base	Base D1		LoRA P0		LoRA D1	
	CR	EX	CR	EX	CR	EX
Llama-3.1-8B	0.525	0.605	**0.733**	0.666	0.722	**0.670**
Qwen3-8B	0.500	0.616	**0.710**	0.655	0.705	**0.662**
Qwen3.5-9B	0.642	0.661	0.703	0.652	**0.717**	**0.673**
Qwen3-32B	0.640	0.661	0.707	0.663	**0.730**	**0.687**
Mean	0.577	0.636	0.713	0.659	**0.718**	**0.673**

Table 8: Per-base LoRA macro-F1, 3-seed means. **Base D1**: zero-shot prompted, no LoRA. **LoRA P0**: adapter evaluated with the bare training prompt. **LoRA D1**: same adapter, D1 prompt swapped in at inference time. **Bold** marks each row's best state per setting.

Method	TCFD-CR	TCFD-EX
ClimateBERT-TCFD (supervised)	0.710	0.648
Zero-shot + definitions (best D)	0.657 (GPT-5-mini, D1)	0.707 (GPT-4o, D3)
Random few-shot (k=8, n=2)	0.689 (Qwen3.5-27B)	0.705 (Qwen3.5-27B)
TF-IDF retrieval (best-on-CR)	0.718 (Llama-3.3-70B)	**0.713** (Qwen3.5-27B)
Dense retrieval (best-on-CR)	0.714 (Qwen3.5-9B)	0.689 (Qwen3.5-27B)
LoRA + P0 (training prompt)	**0.733** (Llama-3.1-8B)	0.666 (Llama-3.1-8B)
LoRA + D1 (D1 swapped in)	0.730 (Qwen3-32B)	0.687 (Qwen3-32B)

Table 9: Best macro-F1 per strategy on TCFD-CR (in-source) and TCFD-EX (cross-source); champion model in parentheses. **Bold** marks the overall champion per setting.

Controlled effect	In-source	Cross-source	Cross − in (95% CI)
Similar correct-class examples	+16.7	+9.5	-7.3 [$-12.8, -1.9$]
Similar other-class examples	-9.2	-10.2	-1.0 [$-5.6, +3.8$]
Full similarity retrieval	+7.6	-0.7	-8.3 [$-15.4, -1.3$]

Table 10: Accuracy-point effects from the balanced intervention, averaged over two models (Qwen3-8B and Llama-3.1-8B) and two retrievers (TF-IDF and dense) on 112 targets (14 per source and class). Each prompt contains eight examples, two per class.